

# Masterclass on nonnegative Low-Rank Approximations

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Lecture 1 — Introduction and matrix low-rank models, 5-9 October 2026, Univ. Trento

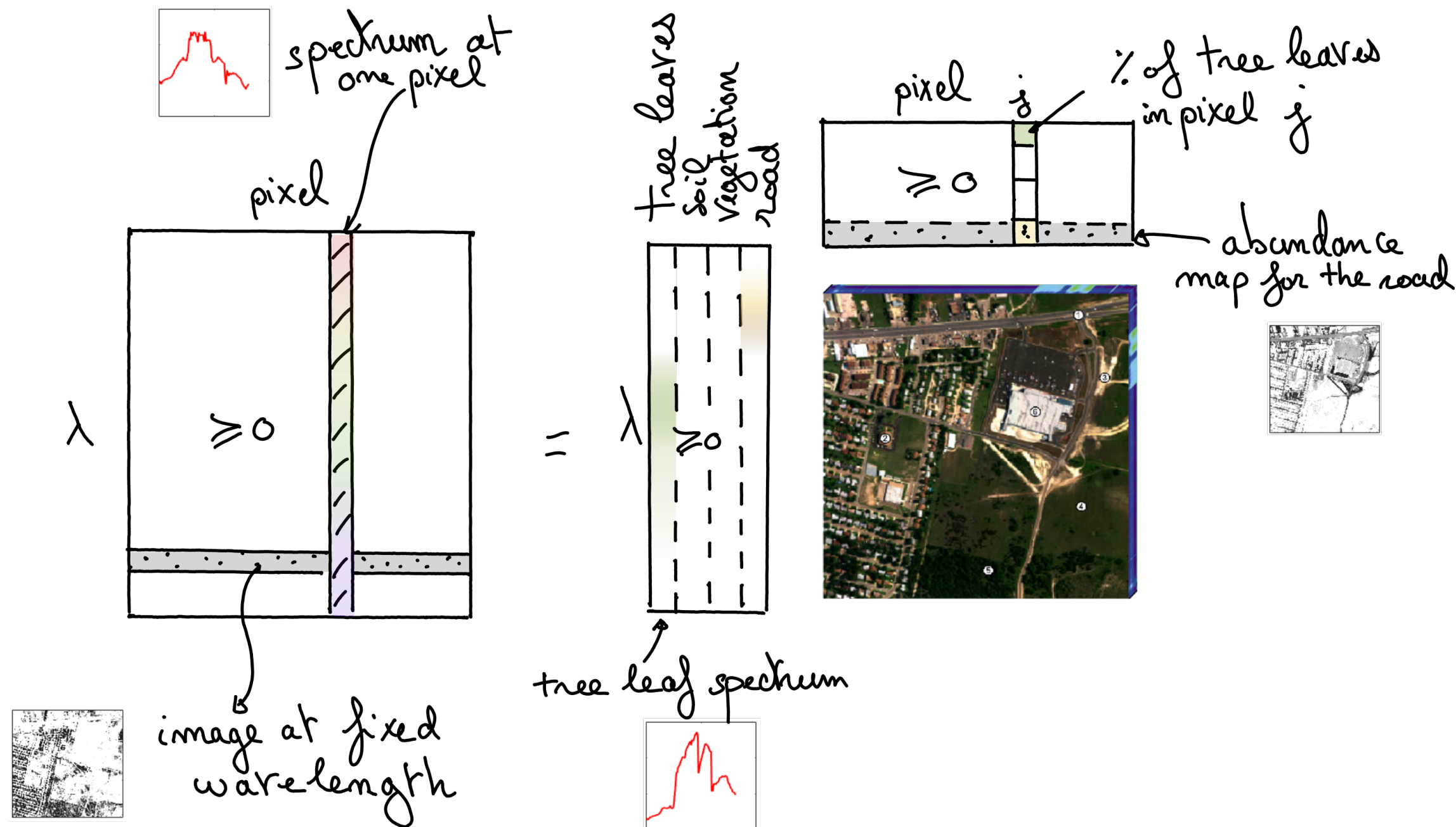
Jérémy E. Cohen  
CNRS, CREATIS, Lyon

2026-10-01

# Introduction: Regularized Low-Rank Approximations

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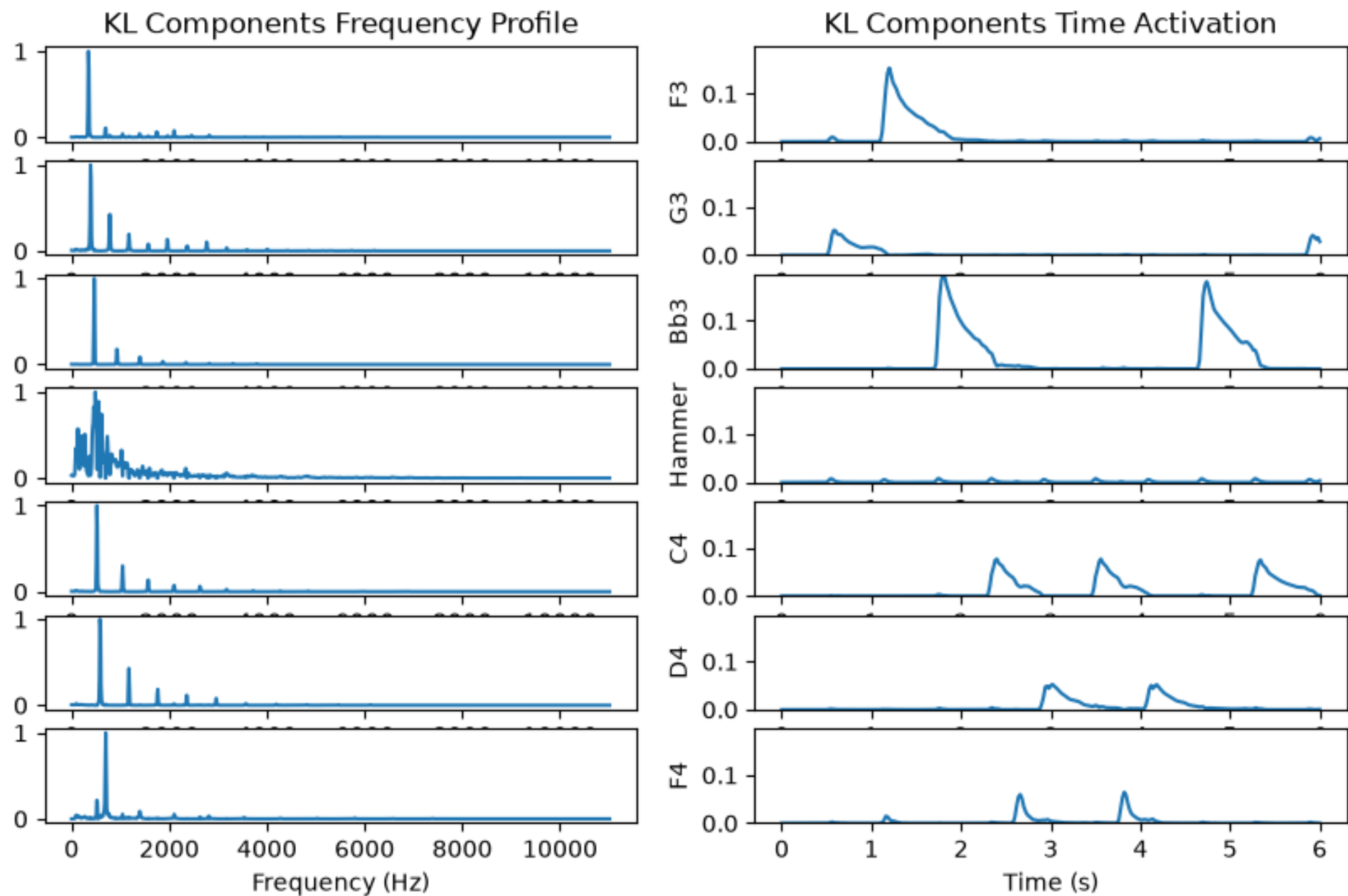
# Example NMF with hyperspectral imaging







# Example NMF for piano transcription (Lecture 4)



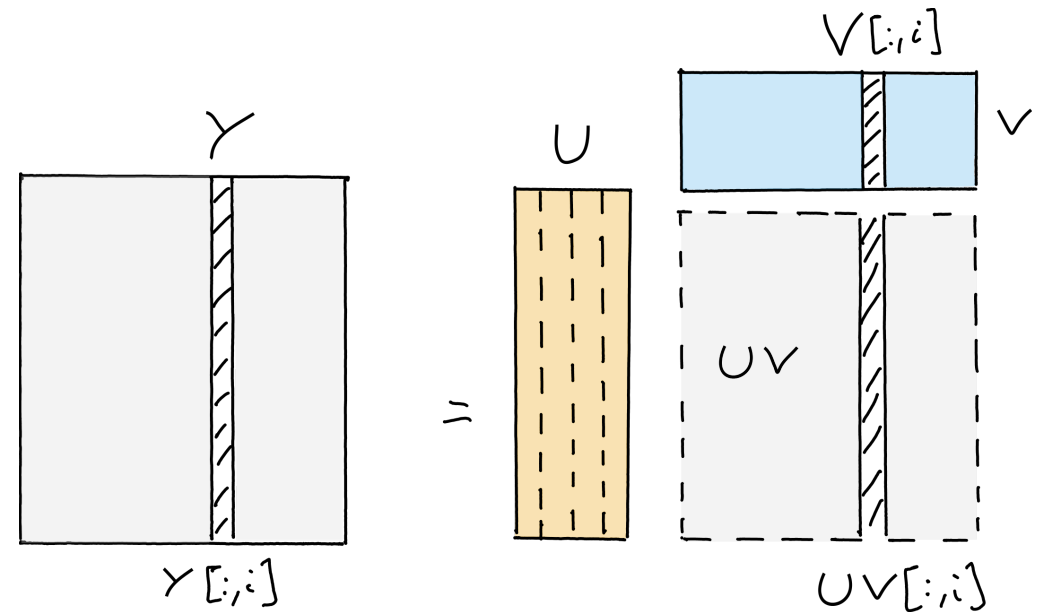


# Low-rank approximations

- **Workhorse methods** in unsupervised ML:
  - Dimensionality reduction [Tucker1966], [Golub1989], [Hotelling1992]
  - Blind source separation [Harshman1970], [Paatero1997], [Lee1999]
- **Formal LRA formulation:**

$$\min_{U \in \mathbb{R}^{m \times r}, V \in \mathbb{R}^{n \times r}} f(Y, UV^T)$$

- Rank  $r \ll m, n$
- Typical loss function:  $f(Y, X) = \|Y - X\|_F^2$



# Importance of physical meaning in LRA

- Physical Interpretability:

- Factors  $U$  and  $V$  often represent physical quantities (e.g., source spectra & mixing weights)
- Reconstruction of ground-truth  $(U^*, V^*)$  is an **inverse problem**
- Requires **uniqueness** for correct physical interpretation

- Indeterminacy of Unconstrained LRA:

- Solution is never unique without constraints:

$$UV^T = (UP)(P^{-1}V^T)$$

for any invertible matrix  $P \in \mathbb{R}^{r \times r}$

- Regularized LRA (rLRA):

$$\min_{U,V} f(Y, UV) + g_U(U) + g_V(V)$$

- $g_U, g_V$  promote prior properties (smoothness, sparsity, nonnegativity)
- Restricts solution space to promote uniqueness

## Independent Factor Regularization

Regularizations must apply to  $U$  and  $V$  **independently**. A joint penalty  $g(UV^T)$  cannot discriminate between rotated solutions  $UP(VP)^T$ .

1 Essential uniqueness is the uniqueness up to permutations and scaling ambiguities inherent to LRA models.

# Regularized LRA

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## PCA (Orthogonal LRA)

- **Constraint:** Imposes orthogonality  
 $U^T U = I, V^T V = I$
- **Uniqueness:** Essentially unique factors (if singular values are distinct)
- **Limitation:** Orthogonality is often **not physically meaningful**
- **Impact:** Estimated factors  $(U, V)$  can be poor approximations of ground truth  $(U^*, V^*)$

## NMF (Nonnegative LRA)

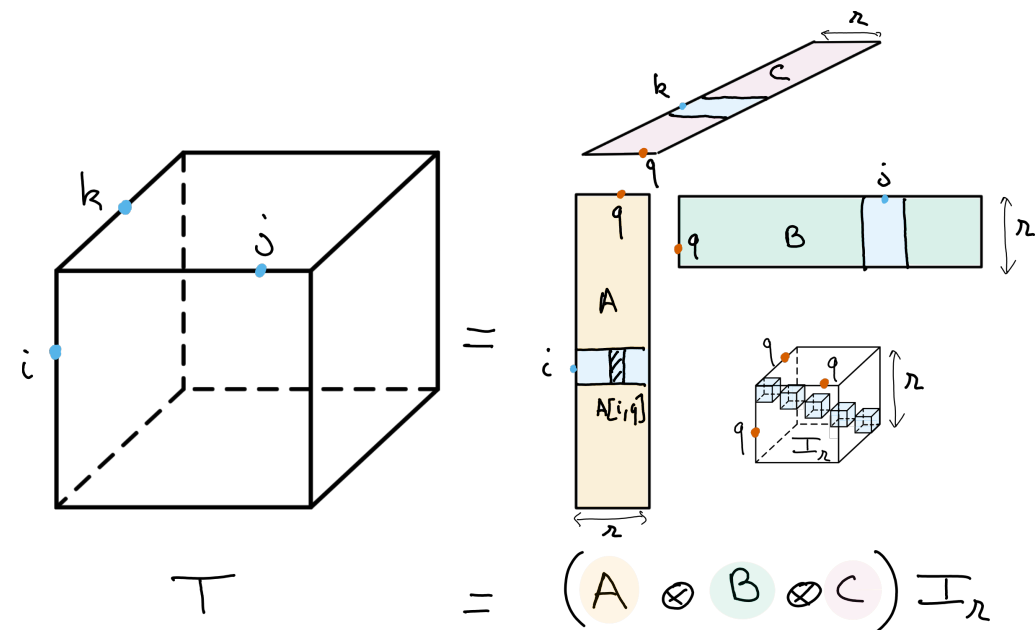
- **Constraint:**  $g_U, g_V$  set to indicator functions of  $\mathbb{R}_+$
- **Uniqueness:** Unique under realistic conditions [Gillis2020]
- **Advantage:** Elementwise nonnegativity is a **natural physical prior**
- **Use Cases:** Ideal for source separation, pattern mining, spectral unmixing

Lecture 1: rLRA theory

Lecture 4: NMF for music transcription

# Tensor LRA

- **Multi-way Data Extension:**
  - Extends matrix LRA to data with  $> 2$  dimensions (tensors)
- **Canonical Polyadic Decomposition (CPD):**
  - Multilinear diagonalization into  $r$  rank-one factor matrices  $(A, B, C, \dots)$
- **Inherent Identifiability:**
  - Uniquely identifiable **without regularizations** under mild conditions [Kruskal1977]
- **Role of Regularization in Tensor LRA:**
  - Nonnegativity & sparsity still essential in practice
  - Enhances interpretability and robustness against noise



Lecture 1: rLRA theory

Lecture 5: Nonnegative Tucker decomposition for music segmentation

# Optimization and LRA

- Optimization Challenges:

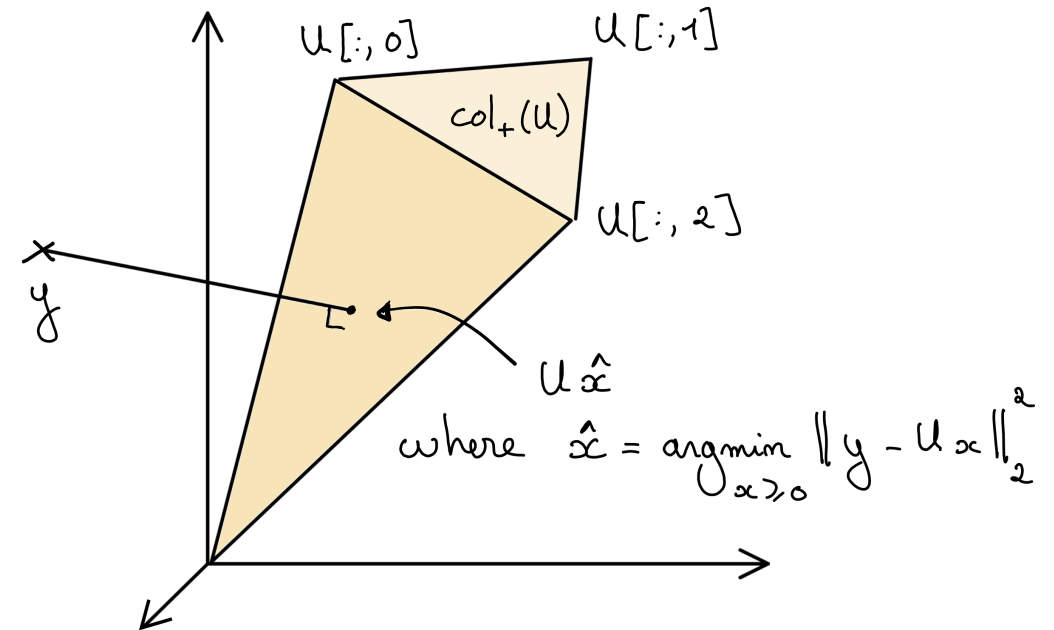
- Nonconvex due to factor products ( $UV^T$ )
- Often nonsmooth (e.g., nonnegativity constraints)
- Multi-block optimization landscape

- Block Convexity & AO:

- Cost is often convex w.r.t. **one factor** while fixing others
- **Nonnegative Least Squares (NNLS):**

$$\min_{U \in \mathbb{R}_+^{m \times r}} \|Y - UV^T\|_F^2$$

- Leverages convex numerical optimization in **Alternating Optimization (AO)** strategies



Lecture 2: NNLS algorithms theory

Lecture 3: Implementation of NNLS and NMF algorithms

# Challenges in rLRA

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- **Numerical Optimization:**
  - Nonconvex, multi-block, and nonsmooth
- **Integrating Side Information:**
  - Template libraries, downstream tasks, deep priors, sparse bases
- **Multimodality & Coupled rLRA:**
  - Joint processing across diverse sensors (e.g., EEG + fMRI)
- **Model Properties & Identifiability:**
  - Elusive properties (e.g., Nonnegative Tucker [Saha2025])
  - Practical, verifiable guarantees are hard to obtain
- **Scientific Software Ecosystem:**
  - Tensor software is fragmented ( $> 70$  packages [Psarras2022])
  - Lack of large-scale contraction support & speed/memory trade-off caching
- My HDR manuscript that contains the long version of this masterclass [\[link\]](#). Some other books to check:
  - Book chapter with Pierre Comon and Rasmus Bro.
  - The book of Golub and Van Loan provides a complete description of matrix low-rank models.
  - The recent book of Grey Ballard and Tammy Kolda provides a deeper introduction to tensor decompositions, with a focus on computational aspects.



# First Lecture: regularized LRA

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# Matrix low-rank factorizations: linear dimensionality reduction

- **Subspace View:**

- $M \in \mathbb{R}^{n_1 \times n_2}$ : collection of  $n_2$  vectors in  $\mathbb{R}^{n_1}$
- Data lies in a subspace of dimension  $r \leq \min(n_1, n_2)$
- Compact representation for data compression & unsupervised feature mining

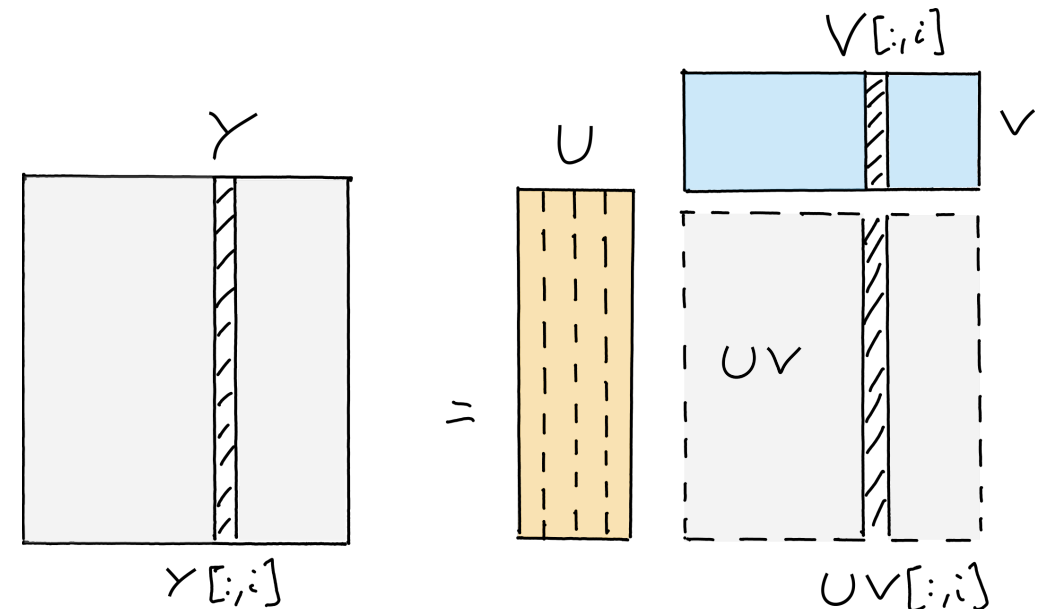
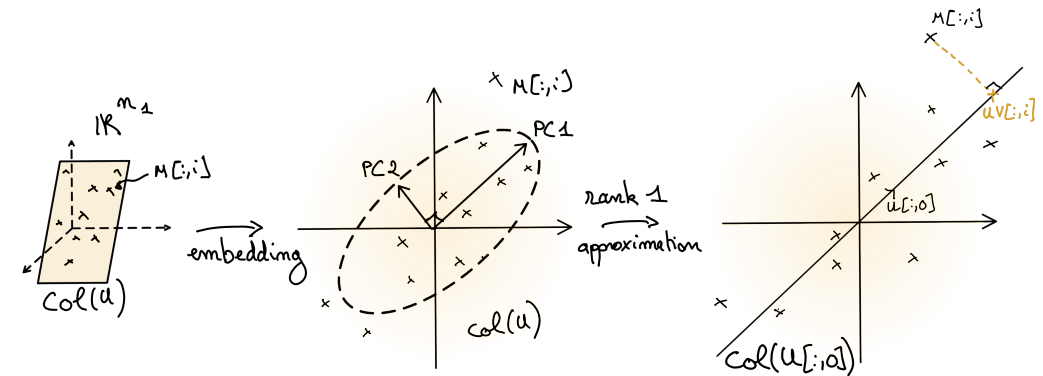
- **Mathematical Model:**

$$\forall j \leq n_2, \quad M[:, j] = UV^T[:, j]$$

- $U \in \mathbb{R}^{n_1 \times r}$ : basis of the subspace  $\text{col}(M)$
- $V^T[:, j] \in \mathbb{R}^r$ : coordinates in basis  $U$
- Minimal  $r$  is the **rank**  $\text{rank}(M)$

- **Computation & Link to SVD / PCA:**

- Computed via **Singular Value Decomposition (SVD)**
- Centered data + truncated SVD  $\implies$  **Principal Component Analysis (PCA)**



# Factorization as a sum of rank-one matrices & atomic decomposition

## Sum of Rank-One Matrices

- Formulation:

$$M = \sum_{q=1}^r U[:, q] V^T[q, :]$$

$$M[i, j] = \sum_{q=1}^r U[i, q] V[j, q]$$

- Both  $U \in \mathbb{R}^{n_1 \times r}$  and  $V \in \mathbb{R}^{n_2 \times r}$  play symmetric roles
- Compresses  $n_1 n_2$  entries into  $(n_1 + n_2)r$  parameters
- Naturally extends to **tensor decompositions**

## Atomic Decomposition View

- Formulation:

$$M = \sum_{q=1}^r M_q, \text{ rank}(M_q) \leq 1$$

- Atoms are rank-one matrices

$$M_q \in \{ab^T \mid a \in \mathbb{R}^{n_1}, b \in \mathbb{R}^{n_2}, a, b \neq 0\}$$

- Rank measures matrix complexity
- Connects LRA directly to **inverse problems**

Important

$\text{rank}(M)$  = minimal number of rank-one terms

# Why are data matrices often (approximately) low-rank?

## 1. Latent Variable Models:

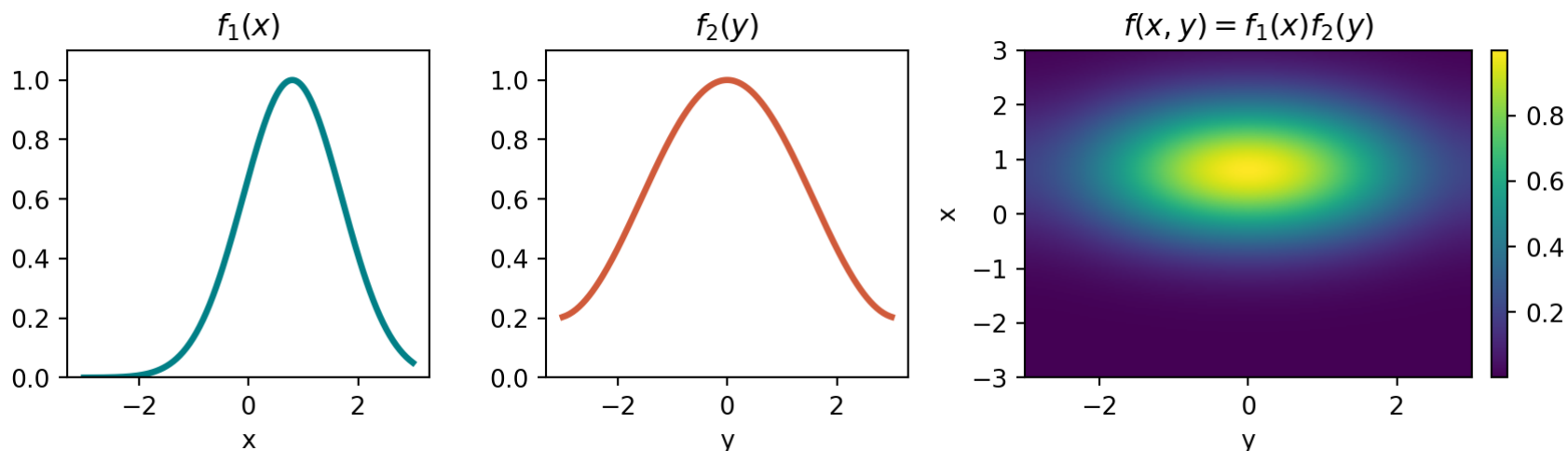
- Generative models produce low-rank matrices [Udell2019]

## 2. Separable Physical Maps:

- $f(x, y) = f_1(x)f_2(y)$  ubiquitous in physics
- Think of quantum particle interactions, Beer-Lambert law, etc.

## 3. Blind Source Separation:

- $M = UV^T$ :  $U$  = templates,  $V$  = mixing coefficients
- Think linear mixing model in hyperspectral imaging, audio source separation (see Lecture 4), etc.
- Also many sensing devices are approximately linear (MRI, scanners and such)
- Key notion: linear combination of atoms (columns of  $U$ )



# Low-rank approximations: formulation & SVD

In the presence of noise, we are interested in low-rank approximations of a full-rank matrix.

## Problem Formulations

- **Matrix Approximation:**

$$\min_{\text{rank}(N) \leq r} \|M - N\|_F^2$$

where  $\|M - N\|_F^2 = \sum_{i,j} (M[i, j] - N[i, j])^2$

- **Burer-Montero Factorization:**

$$\min_{U \in \mathbb{R}^{n_1 \times r}, V \in \mathbb{R}^{n_2 \times r}} \|M - UV^T\|_F^2$$

## The Eckart-Young Theorem

Eckart-Young Theorem (1936)

If  $M = ASB^T$  is the SVD of  $M$ , then the truncated SVD:

$$N^* = A[:, : r] S[:, : r] B^T[:, : r]$$

is optimal in Frobenius (and spectral) norm.

- Residual Error:  $\sum_{q=r+1}^{\min(n_1, n_2)} S[q, q]^2$
- Unique if  $r$ -th singular value is distinct
- Remark: one of the few non-convex problems with a closed-form global solution

# Identifiability & Robustness

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- Identifiability?

- No!  $M = U_1 V_1^T = U_2 V_2^T \implies U_1 = U_2 Q$  for invertible  $Q \in \mathbb{R}^{r \times r}$
- Rotation ambiguity requires additional constraints/regularization

- Robustness to Noise?

- Directly governed by the **condition number** of  $M$

## Important

Regularized LRA = LRA + constraints/priors to promote uniqueness.

# Nonnegative Matrix Factorization

- Definition & Model:

$$\text{Find } W \in \mathbb{R}_+^{n_1 \times r}, H \in \mathbb{R}_+^{n_2 \times r} \text{ s.t. } M \approx WH^T$$

- Physical Motivation:

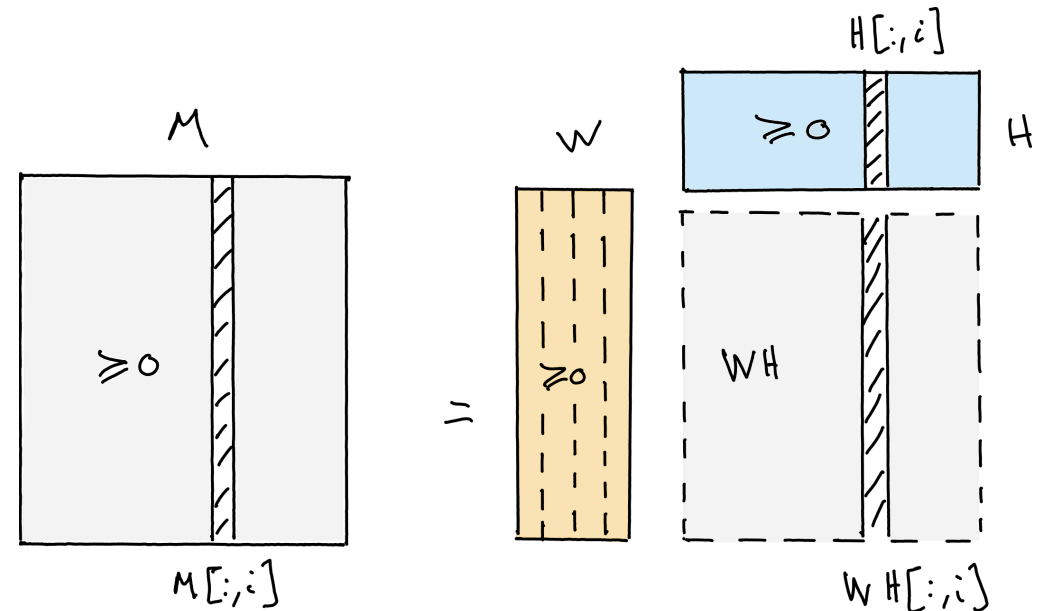
- Nonnegativity represents concentrations, spectra, probabilities, power, etc.

- Fitting NMF:

- NP-hard problem [Vavasis2010]
- Solved via non-convex optimization (e.g., HALS, MU, CCD)

- Rank Selection:

- Truncated SVD deflation **does not apply** to NMF
- Rank  $r$  must be pre-specified or tuned



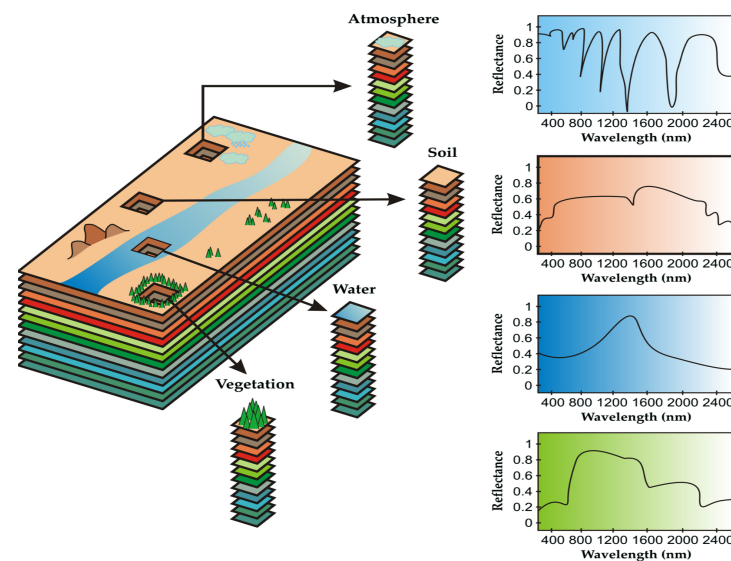
# History of NMF

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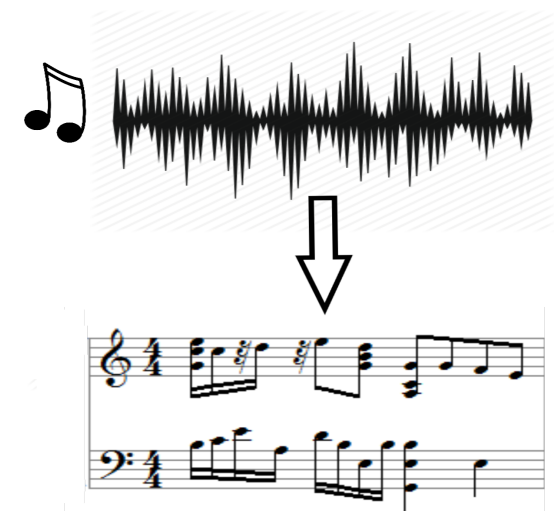
- **Early ideas:** nonnegative matrix factorization is rooted in older work on positive matrix decompositions and chemometrics, where nonnegativity was a natural physical assumption.
- **1990s:** the modern NMF formulation emerged through the connection with source separation and latent-variable models; Lee and Seung popularized the problem in the machine learning community. [Lee & Seung1999]
- **2000s:** NMF became a standard tool for document clustering, hyperspectral unmixing, and image analysis, with the famous multiplicative updates and the rise of interpretable factor models.
- **2010s–present:** the field broadened to regularized NMF, sparse/volume-constrained variants, separable NMF, and tensor extensions, driven by identifiability and scientific applications.
- NMF is now seen as a canonical example of a regularized low-rank model with strong physical meaning.



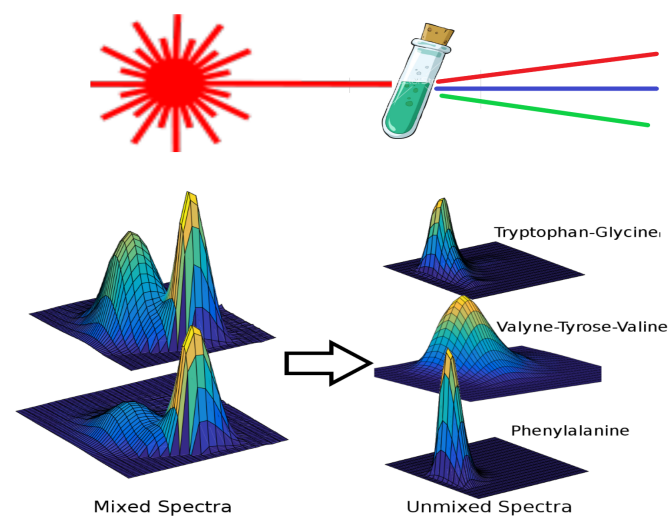
# Applications of NMF



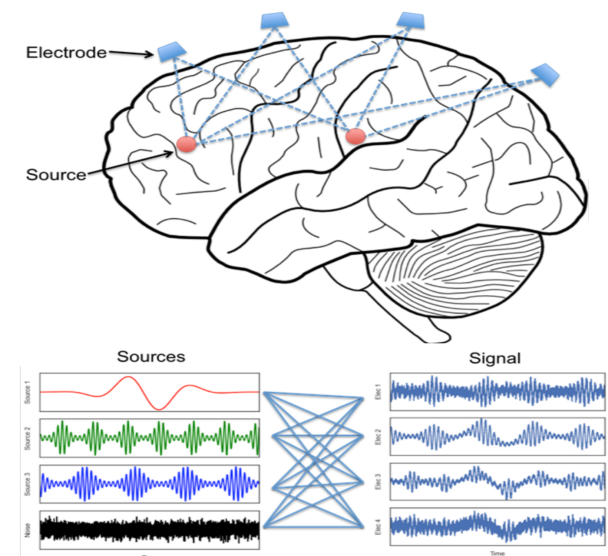
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AUDIO



CHIMIOMETRIE



NEUROSCIENCES



# Geometric interpretation and identifiability of exact NMF

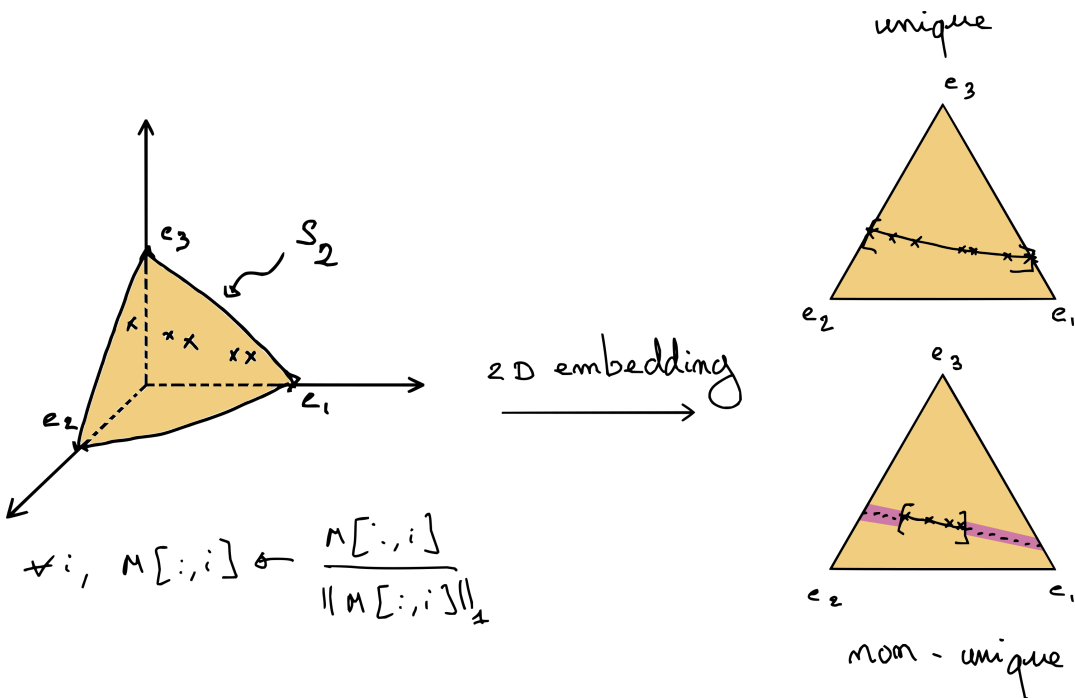
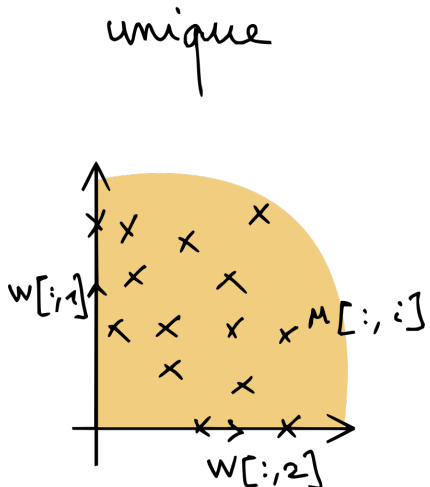
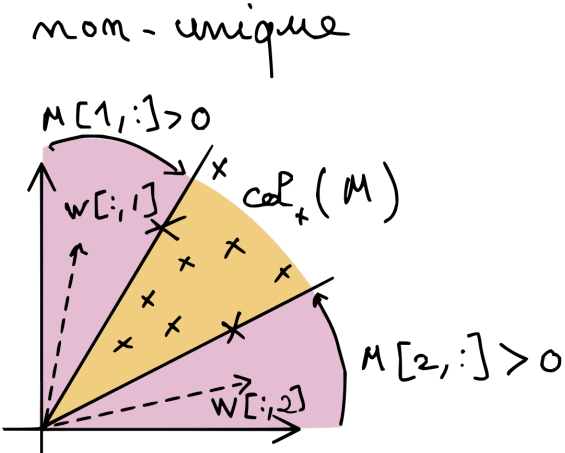
The columns of  $M$  must lie inside the cone generated by the columns of  $W$ :

$$\mathcal{C}(W) = \{Wx : x \geq 0\}.$$

Different cones may cover the same data, so exact NMF is not generally unique.

## Simplex View & Identifiability

- Projection onto the unit simplex (e.g.  $\ell_1$  normalization)
- Data lives in a polytope inside the simplex
- **Uniqueness condition:** Data points must touch the boundaries of the simplex
- **Sufficiently Scattered Condition:** Guarantees uniqueness if columns of  $H^T, W$  spread widely [Gillis2020]



# Regularized NMF: separable NMF

- Pure-Pixel / Pure-Data Assumption:

- Columns of  $W$  are selected directly from columns of  $M$ :

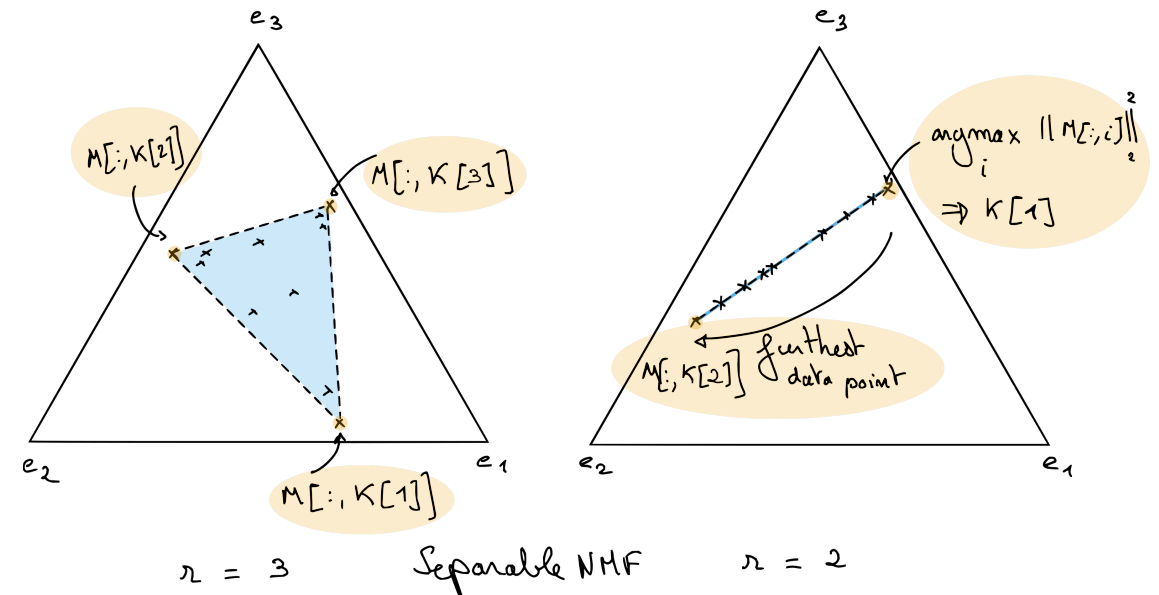
$$M = M[:, \mathcal{K}] H^T \quad \text{with } |\mathcal{K}| = r$$

- Advantages:

- Drastically restricts solution space
- Solvable in **polynomial time** (e.g., Successive Nonnegative Projection Algorithm - SNPA [Gillis2014])
- Excellent initialization for other algorithms
- High interpretability (extracts real data templates)

- Limitation:

- Fails if no “pure” data points exist (all data points are mixtures)
- Sensitive to noise and outliers



## Note

- Rank-two exact separable NMF can be computed in closed form
- Rank-one exact separable NMF is computed through SVD by the Perron-Frobenius theorem

# Sparse and Minimum-Volume NMF

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## Sparse NMF

- Formulation ( $\ell_1$  regularization):

$$\min_{W \geq 0, H \geq 0} \|Y - WH^T\|_F^2 + \lambda (\|W\|_F^2 + \|H\|_1)$$

- Mechanism & Benefit:

- Pushes entries in  $W$  and/or  $H$  towards zero
- Drives abundance/coefficients  $H$  towards the **simplex boundary**
- **Restores uniqueness** when standard NMF is unidentifiable

## Minimum-Volume NMF

- Concept:

- Fits cone  $\mathcal{C}(W)$  tightly around the data hull
- Relaxes the separability assumption (works without pure pixels)

- Objective:

- Minimizes volume of  $W$ , e.g., via  $\log \det(W^T W)$
- Maximizes volume of abundances  $H^T$

# Fitting rLRA & NMF: Optimization Frameworks

## Optimization Paradigms

- **Alternating Optimization (AO):**
  - *Block Coordinate Descent*: updates one factor while keeping others fixed
  - Transforms non-convex joint problem into a sequence of **convex subproblems**
  - Most widely used framework for matrix & tensor rLRA
- **All-at-Once Optimization:**
  - Joint updates of  $(W, H)$  simultaneously
  - Gradient descent, Quasi-Newton, projected gradient methods

## The Core Subproblem: NNLS

- **Nonnegative Least Squares (NNLS):**
  - Fixing  $H$  yields a convex subproblem in  $W$ :
$$\min_{W \geq 0} \|M - WH^T\|_F^2$$
  - Fundamental building block for NMF & nonnegative tensor factorizations

Coming up in Lecture 2

The **NNLS subproblem** and its algorithms (HALS, MU, Active Set, PG) will be covered in detail in **Lecture 2**.

# Software Ecosystem for NMF

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## Scikit-Learn

- Industry standard in Python
- Implements Multiplicative Updates (MU) & Coordinate Descent (CCD)
- Supports sparsity constraints ( $\ell_1/\ell_2$ )
- Fast & well-maintained

## Nimfa

- Dedicated NMF toolbox in Python
- Implements many NMF variants (Bayesian, graph-regularized, etc.)
- Highly flexible
- Unmaintained since 2019

## TensorLy

- Multi-way array library
- Implements MU and HALS (Hierarchical Alternating Least Squares)
- Generic NNLS solver building blocks
- CPU/GPU backends (NumPy, PyTorch, CuPy)

# Simulation: Standard NMF vs Ground Truth

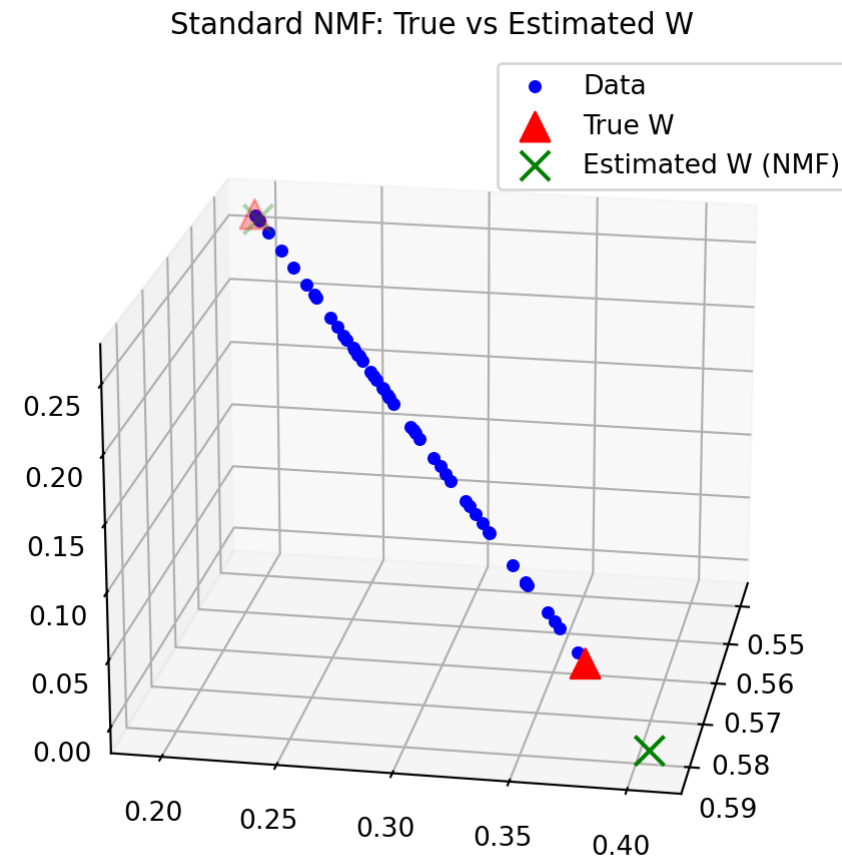
- Rank-2 Exact NMF Data:

- $M = W^* H^{*T}$  ( $n_1 = 3, n_2 = 50, r = 2$ )

- Standard NMF Output:

- Low fitting error, but **fails to recover true  $W$**  due to non-uniqueness

```
1 import numpy as np
2 from tensorly.decomposition import non_negative_parafac_hals
3
4 rng = np.random.default_rng(5)
5 W_true = rng.random((3, 2))
6 H_true = rng.random((50, 2))
7 M = W_true @ H_true.T
8
9 # Fit NMF with TensorLy (HALS)
10 out = non_negative_parafac_hals(M, 2, init="svd", n_iter_max=100)
11 We = out[1][0]
```

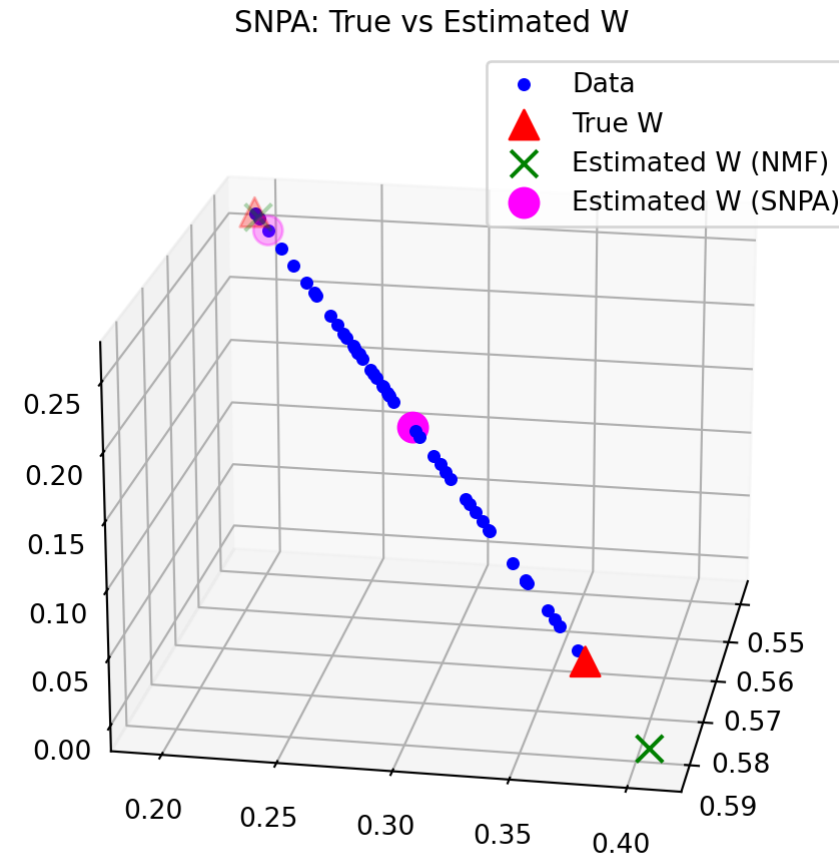




# Simulation: Recovering Ground Truth with Separable NMF (SNPA)

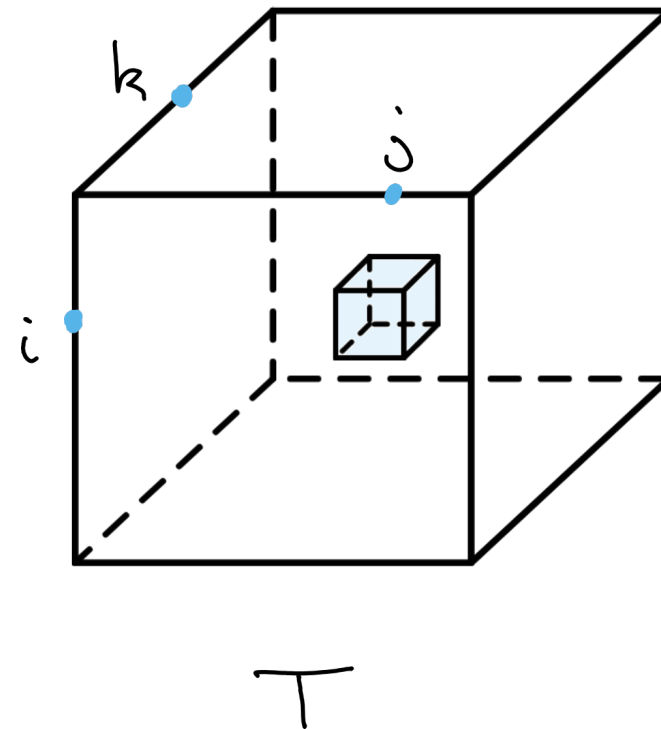
- Separable NMF (SNPA):
  - Leverages pure-data assumption
  - Identifies extreme data vectors
- **Result:** Successfully recovers ground truth  $W$ !

```
1 from tensorly_hdr.sep_nmf import snpa
2
3 # Compute Separable NMF via SNPA
4 _, We_snpa, _ = snpa(M, 2)
```



# Tensors

- **Multiway arrays:** tensors extend matrices to more than two modes
  - Matrix:  $M[i, j]$
  - Third-order tensor:  $T[i, j, k]$
- **Modes** are the different dimensions of a tensor
- **Order  $d$**  is the number of modes
- Example:  $T \in \mathbb{R}^{n_1 \times n_2 \times n_3}$  is a third-order tensor



# Tensors: several compatible viewpoints

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- **Multiway array:** a natural representation for data, tensors are elements of  $\mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$ .
- **Tensor-space vector:** an element of

$$E_1 \otimes \dots \otimes E_d$$

where  $E_i$  are vector spaces such as  $\mathbb{R}^{n_i}$ .

- **Multilinear map or form:** a higher-order generalization of matrices as linear operators.
- **Higher-order moments:** common in statistics and machine learning

## Why ??

- Different fields use different viewpoints
- The viewpoints are mathematically compatible, but emphasize different operations
- For tensor LRA, start from the matrix case and generalize
- A tensor-specific challenge: efficient high-order contractions [Bientinesi2022]

# The tensor product

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## From multilinear to linear

- Given vector spaces  $E$  and  $F$ , the tensor product combines  $x \in E$  and  $y \in F$  into  $x \otimes y$
- It creates a space that contains interactions between the two spaces
- Every bilinear map  $u$  can be represented by a linear map  $w_u$ :

$$u(x, y) = w_u(x \otimes y)$$

- This is the **universality property** of the tensor product [Schwartz1975]

## Tensor-product space

- The tensor-product space is generated by products  $e_i \otimes f_j$  where  $\{e_i\}_i$  and  $\{f_j\}_j$  are bases of  $E$  and  $F$ , respectively.
- In finite dimensions:

$$\dim(E \otimes F) = \dim(E) \dim(F)$$

- The construction generalizes to any order  $d$ :

$$E_1 \otimes \cdots \otimes E_d$$

# Tensor products in practice

## Outer product representation

- Identify  $E$  and  $F$  with Euclidean spaces
- The tensor product becomes the outer product:

$$(x \otimes y)[i, j] = x[i]y[j]$$

$$x \otimes y = xy^T$$

- For instance, for matrices, the linear map associated with the inner product is the trace:

$$\langle x, y \rangle = \text{Tr}(xy^T)$$

## Operators and Kronecker products

- Linear operators acting on tensor spaces form tensor-product spaces too [Hackbusch2012]
- The Kronecker product builds this tensor space of operators and acts modewise:

$$(A \otimes B)(x \otimes y) = Ax \otimes By$$

### Tip

It is possible to perform matrix and tensor LRA with the Kronecker product instead of the outer product!

# Rank-one tensors are generators of the full tensor space

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## Rank-one tensors

- A tensor of the form

$$x \otimes y$$

is called **rank one**

- Rank-one tensors are the elementary building blocks of tensor spaces
- Because tensor spaces are vector spaces, finite sums remain tensors:

$$T = \sum_{q=1}^p x_q \otimes y_q$$

## Generating the tensor space

- In finite dimensions, products of basis vectors generate the whole space:

$$\{e_i \otimes f_j\}_{i,j} \text{ is a basis of } E \otimes F$$

- Therefore, every tensor can be expressed as a finite sum of rank-one tensors
- This is the tensor analogue of expanding a matrix in the canonical basis

# Rank-one tensors and tensor rank

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## Not every tensor is rank one

- A sum of rank-one tensors is generally not rank one
- Matrix example:

$$e_1 f_1^T + e_2 f_2^T$$

- This matrix cannot be written as a single outer product  $xy^T$
- The same phenomenon occurs in higher-order tensor spaces

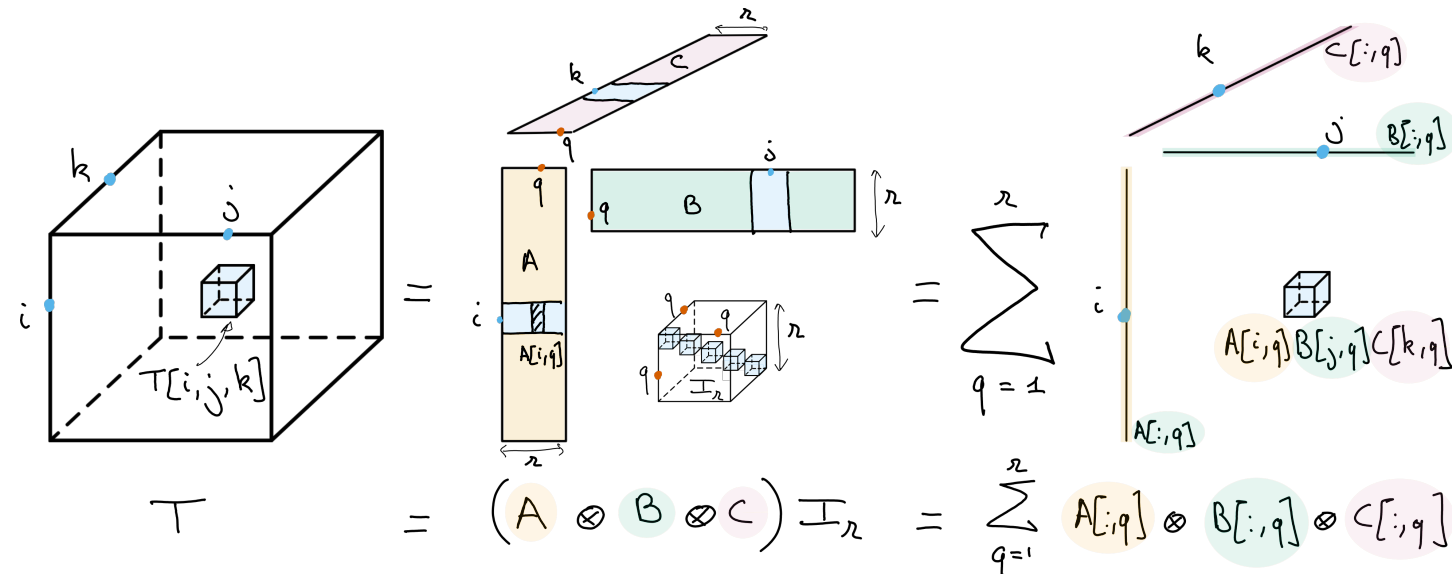
## Tensor rank

- **Tensor rank** is the smallest number of rank-one terms needed:

$$\text{rank}(T) = \min \left\{ p : T = \sum_{q=1}^p x_q \otimes y_q \right\}$$

- For matrices, this recovers the usual matrix rank
- For higher-order tensors, this definition leads naturally to the CP decomposition

# Canonical Polyadic Decomposition



- CPD / PARAFAC represents a tensor as a sum of rank-one tensors
- Factor matrices  $A = [a_1, \dots, a_r]$ ,  $B$ , and  $C$  collect the rank-one components

- Kruskal notation:

$$T = \llbracket A, B, C \rrbracket$$

- Equivalent operator notation:

$$T = (A \otimes B \otimes C) I_r$$

$$T = A \times_1 B \times_2 C \times_3 I_r$$



# CP components and their ambiguities

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For a third-order tensor, the CP model expands entrywise to

$$T[i, j, k] = \sum_{q=1}^r A[i, q] B[j, q] C[k, q].$$

Each component contributes one separable pattern across all three modes. It can be rescaled without changing the tensor:

$$a_q \otimes b_q \otimes c_q = (\alpha a_q) \otimes (\beta b_q) \otimes ((\alpha\beta)^{-1} c_q).$$

Components may also be permuted, so uniqueness is understood up to scaling and permutation.

# Matrix versus tensor case

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- For matrices, CPD reduces to an exact factorization  $M = XY^T$
- Matrix factorizations are never unique and can be computed efficiently with SVD
- For tensors of order greater than two, CPD can be identifiable under mild assumptions [Kruskal1977]

## Why CPD matters

- A unique decomposition can make factors physically interpretable
- CPD may avoid additional regularization for identifiability
- Applications include chemometrics, telecommunications, imaging, and audio [Bro1998] [Sidiropoulos2000] where interpretability matters
- Computation can nevertheless be highly challenging

# Matrix LRA has a rotation ambiguity

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For any invertible  $Q \in \mathbb{R}^{r \times r}$ ,

$$M = AB^T = (AQ)(BQ^{-T})^T.$$

No unique basis for either factor.

## CPD couples the tensor slices

For a third-order tensor, every slice shares the same factors:

$$T[:, :, k] = A \operatorname{Diag}(C[k, :]) B^T, \quad k = 1, \dots, n_3.$$

The shared factors couple the slices, imposing more structure than a separate matrix factorization of each slice.

## Two slices can reveal the factors

Under suitable full-rank and distinct-eigenvalue assumptions, for two slices,

$$T[:, :, 1]T[:, :, 2]^\dagger = A \operatorname{Diag}\left(\frac{C[:, 1]}{C[:, 2]}\right) A^\dagger.$$

Distinct diagonal entries can determine the columns of  $A$  up to scaling and permutation. This illustrates why tensor structure can make CPD identifiable.

# Fitting CP with alternating least squares

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Given a tensor  $T$ , choose the rank  $r$  and fit a sum of  $r$  rank-one terms:

$$\min_{A,B,C} \|T - \llbracket A, B, C \rrbracket\|_F^2.$$

The objective is nonconvex jointly, but quadratic in each factor when the other two are fixed. Alternating Least Squares (ALS) cycles through these block updates:

- Each block update is a least-squares problem.

$$\min_A \|T_{[1]} - A(B \odot C)^T\|_F^2 = T_{[1]}(B \odot C)(B^T B * C^T C)^{-1}$$

where  $\odot$  is the so-called Khatri-Rao product and  $T_{[1]}$  is the mode-1 unfolding of  $T$ .

- Update  $A$ ,  $B$ , and  $C$  in turn until a stopping criterion is met.
- The high-level method is simple; efficient tensor contractions and caching require care.
- Other approaches include gradient, preconditioned, and second-order methods.

## Approximate CP can be ill-posed

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The set of tensors with rank at most  $r$  is not closed for tensors of order three or higher.

- A best rank- $r$  approximation may therefore fail to exist [Hillar2013].
- ALS can exhibit long periods of slow progress, called swamps.
- Components may grow without bound while canceling in their sum.
- Regularization can improve both well-posedness and interpretability.

# Nonnegative CP decomposition

---

Constrain each factor to be elementwise nonnegative:

$$\min_{A, B, C \geq 0} \|T - \llbracket A, B, C \rrbracket\|_F^2.$$

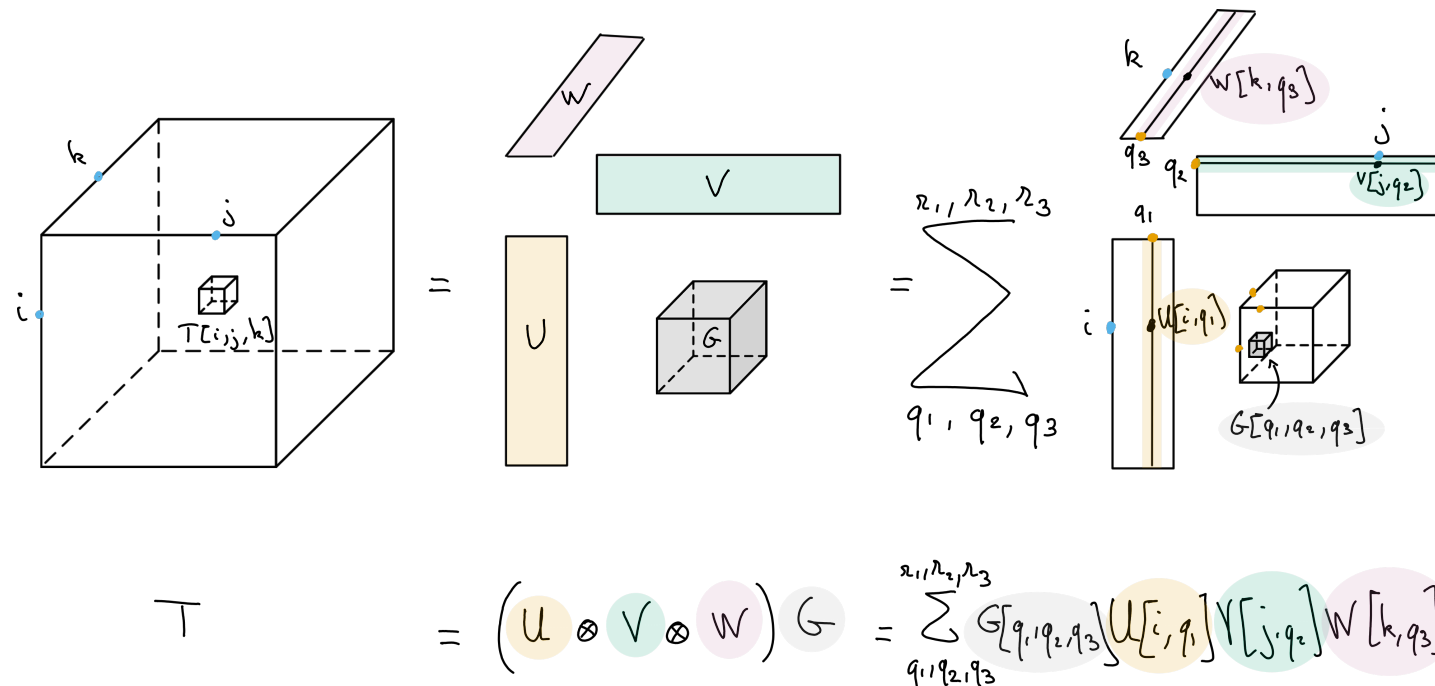
- Alternating updates reduce to nonnegative least-squares subproblems (ANLS).

$$\min_{A \geq 0} \|T_{[1]} - A(B \odot C)^T\|_F^2$$

- Nonnegativity prevents cancellation between components and ensures a best approximation exists.
- Again an NNLS problem, no closed-form solution.
- The NNLS algorithms from Lecture 2 are the optimization building blocks.

# Tucker decomposition compresses each mode

Tucker decomposition projects a tensor onto a lower-dimensional space in every mode and stores interactions in a smaller core tensor.



- Tucker decomposition of a data tensor  $T$ , with multilinear ranks  $r_1$ ,  $r_2$  and  $r_3$ .
- The Tucker decomposition can be viewed as a multilinear dimensionality reduction technique, with orthogonal bases  $U$ ,  $V$  and  $W$  respectively spanning the columns, rows and fibers of the tensor.
- The core tensor  $G$  stores the coefficients into a tensor of size  $r_1 \times r_2 \times r_3$ .

# Tucker model and compression cost

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For a third-order tensor  $T \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ ,

$$T = \llbracket U, V, W; G \rrbracket = G \times_1 U \times_2 V \times_3 W.$$

The model stores

$$n_1 r_1 + n_2 r_2 + n_3 r_3 + r_1 r_2 r_3$$

parameters, which can be much smaller than  $n_1 n_2 n_3$  when the multilinear ranks are small.

## Tucker has rotation ambiguities

- A change of basis in a mode can be absorbed into the core tensor.
- Tucker factors are therefore not unique without further constraints.
- Orthogonality is common; HOSVD provides a standard orthogonal approximation [DeLathauwer2004].
- Tucker is often used as a dimensionality-reduction format rather than an identifiable source model.



## CP and Tucker encode different structure

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### CP

$$T = \sum_{q=1}^r a_q \otimes b_q \otimes c_q$$

- One shared component index across all modes
- No free core tensor
- Rank  $r$  is the number of rank-one terms

### Tucker

$$T = G \times_1 U \times_2 V \times_3 W$$

- A smaller core stores interactions between mode-wise bases
- Separate multilinear ranks  $(r_1, r_2, r_3)$
- Flexible compression, with basis-change ambiguities

# Fitting Tucker with HOSVD

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- Unfold the tensor along each mode to form three matrices.
- Compute an SVD of each unfolding; retain its leading left singular vectors as a mode basis.
- Project the tensor onto these bases to obtain the core.
- Truncating each mode gives a practical multilinear-rank approximation.

# Fitting Nonnegative Tucker

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Impose elementwise nonnegativity on the factor matrices and core, then minimize the reconstruction error:

$$\min_{U, V, W, G \geq 0} \|T - G \times_1 U \times_2 V \times_3 W\|_F^2.$$

- Nonnegative factors can represent additive components such as abundances or intensities.
- With multilinear ranks fixed, alternating updates solve a NNLS subproblem for one block at a time.

$$\min_{U \geq 0} \|T_{[1]} - U G_{[1]} (V \otimes W)^T\|_F^2$$

## Identifiability remains open for Nonnegative Tucker

- Nonnegativity alone does not resolve every Tucker rotation ambiguity.
- Identifiability conditions are more elusive than for CPD and remain an active topic.
- Recent work has made progress, but practical guarantees are still limited [Saha2025].

# Take-home: choose the model and its assumptions

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- Matrix LRA in general is not interpretable but easy to compute.
- Constraints and regularization encode prior knowledge, but their guarantees depend on the model.
- NMF trades a closed-form solution for nonnegative, potentially interpretable factors.
- CPD can be identifiable under suitable conditions; Tucker offers flexible multilinear compression.
- Nonnegative Tucker may be identifiable, but the conditions are still unclear.

Lecture 1: rLRA theory

Lecture 2: NNLS algorithms theory

Lecture 3: NNLS and NMF/NTD algorithms implementation

Lecture 4: Application of NMF to automatic music transcription

Lecture 5: Application of NTD for music structure estimation